

WWW, WIKIPEDIA AND OSNS

AP

NETWORKS OF HUMANS

Theme: no one controls the evolution of the network, which is self-organizing

What is represented (self, news, opinion, concept) and its **lifecycle** determines the structure and the research questions

look at how they connect and when

Direction of communication is important

```
1 import networkx as nx
2
3 eu_DG = nx.DiGraph()
```

GETTING DATA

WWW

- a Networkx *digraph* will represent connectivity
- a companion dictionary maps vertices to URLs of the relative pages
- source: a *scrape* of the 2005 “.eu” domain

X/TWITTER

- today supported by the Python [Tweepy](#) module
- mentions create a network and a *a voting system*.

WIKIPEDIA

- a network of concepts (lemmas/lemmata) maintained by humans (and some bot)
- time-stamped evolution of the network is available [\[dumps.wikimedia.org\]](https://dumps.wikimedia.org)
- contrary to *curated* taxonomies, e.g., [Linnaeus, 1735], this is not a tree

a **directed acyclic graph** is the reference model

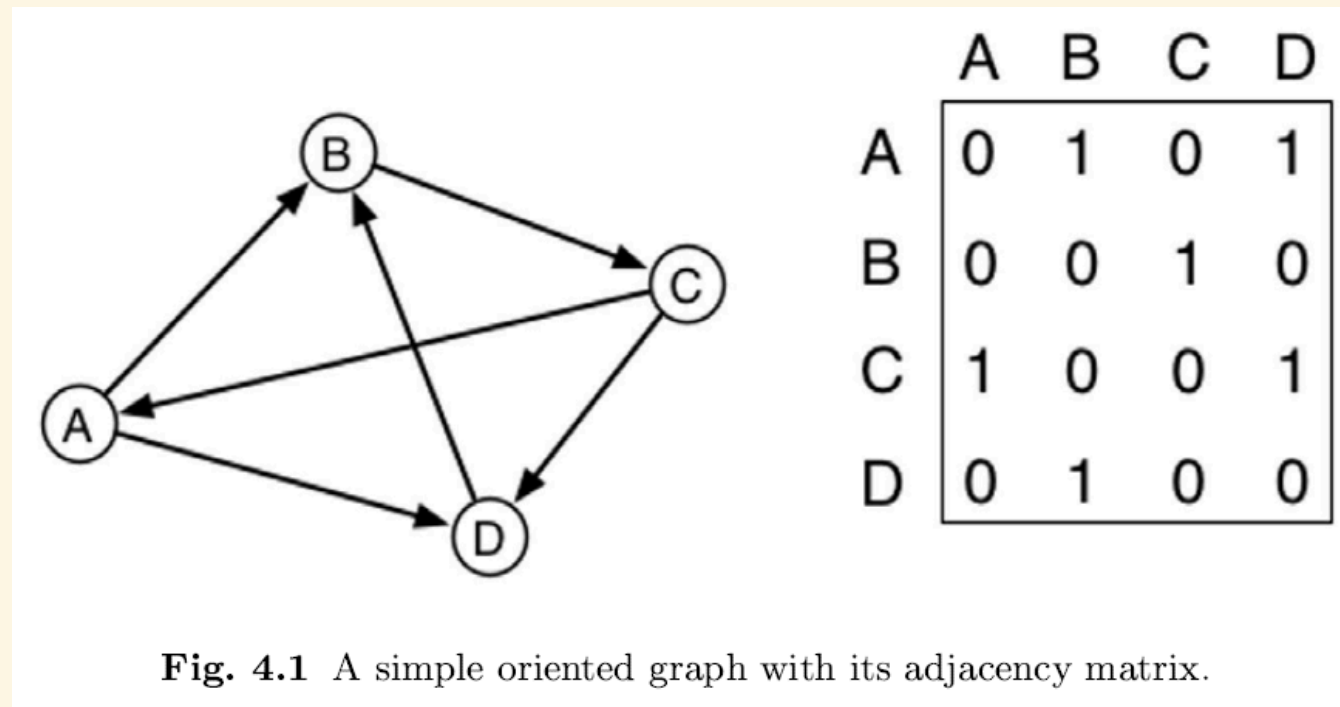
RANKING ALGORITHMS: PAGERANK

PAGERANK IDEA

Rank vertices (page) on the basis of their *importance* for network *navigation*

Importance is captured by the relative value in the dominant Eigenvector of a new matrix, P for 'Page,' that represents *navigation*

Navigation includes a certain prob. of frustration and restart: modeled as a small, uniform prob. to *jump out* to a random node.



PAGERANK AS MATRIX OPERATIONS

A : the directed adjacency matrix of web pages

- it admits *dangling* (or *sink*) pages that link nowhere
- Brin/Page drop internal links

K_0 : has the out-degree of each node on the main diagonal.

$$k_i^{out} = \sum_j A_{ij}.$$

$$K_0 = \text{diag}(k_0^{out}, k_1^{out}, \dots, k_{n-1}^{out}).$$

K_0^{-1} will have $\frac{1}{k_i^{out}}$ on the main diagonal, 0 elsewhere

THE NORMALISED MATRIX

$$N = K_0^{-1} A$$

- similar to A
- all rows sum to 1: they represent the prob. of transitioning ('surfing').

THE ESCAPE MATRIX

$$E = \frac{1}{n} \mathbf{1}\mathbf{1}^T$$

- $\frac{1}{n}$ everywhere, $n = |V|$;
- rows still sum to 1.
- it forces *irreducibility* (see Perron-Frobenius): any two pages are certainly connected
- (current versions may put different values on the main diagonal)

THE PAGE MATRIX

$$P = \alpha N + (1 - \alpha)E$$

Page and Brin empirically set $\alpha = 0.85$

$1 - \alpha = 0.15$: probability of *jumping out* of a navigation *path* and into an arbitrary *restart* node.

Compute the dominant eigenvector $\mathbf{r}_1$ of P .

Perron-Frobenius applies so $\mathbf{r}_1$ will be a ‘proper’ vector of real, positive values.

Compute $\mathbf{r}_1$ by the power method seen earlier:

- iterate the $\mathbf{r}_1^{(t+1)} = P\mathbf{r}_1^{(t)}$ followed by normalisation;
- until it converges to a fixpoint value.

RANKING ALGORITHMS: HITS

HITS IDEA

Hyperlink-Induced Topic Search by [Kleinberg, 1999]

Sees importance of a node in a more nuanced way:

Pages that are important for consultation, e.g., train schedules, have *authority* and tend to be *terminal*

Well-connected *hub* pages that facilitate navigation, e.g., Time Out, are useful but not authoritative per se

1. authority score $\mathbf{au(i)}$

2. hub score $\mathbf{h(i)}$

HITS AS MUTUAL RECURSION

Hub-iness influences authority which in turns influences hub-iness:

$$au(i) \propto \sum_{j \rightarrow i} h(j)$$

page i is authoritative proportionally to the sum of the hub-iness of the pages that link to it.

$$h(i) \propto \sum_{i \rightarrow j} au(j)$$

page i is hub proportionally to the sum of the authoritativeness of pages that it links to.

COMPUTING HITS SCORES

We could start with assigning 1 everywhere and hoping that mutual recursion will converge to stable au and h values.

As with Von Mises' method, we normalise vectors to 1 at each iteration.

LINEAR ALGEBRA DERIVATIONS

$$\mathbf{h} \propto AA^T \mathbf{h} = \lambda_h AA^T \mathbf{h}$$

$$\mathbf{au} \propto A^T A \mathbf{au} = \lambda_{au} A^T A \mathbf{au}$$

I.e., we can find $\mathbf{h}$ and $\mathbf{au}$ separately by solving the eigenvalue problem for the matrices AA^T and $A^T A$

MAIN RESULT

For *primitive* matrices (i.e., connected networks, no dead-ends/sinks)

$$\mathbf{h} \propto AA^T \mathbf{h} = \lambda_h AA^T \mathbf{h}$$

$$\mathbf{au} \propto A^T A \mathbf{au} = \lambda_{au} A^T A \mathbf{au}$$

- convergence is assured;
- dominant λ is unique and
- values for $\mathbf{h}$ and $\mathbf{au}$ will be all positive, as desired.

(negative values have no interpretation here)

COMMUNITY DETECTION

FINDING SOCIAL STRUCTURES

this is an example of Provost-Fawcett's problems

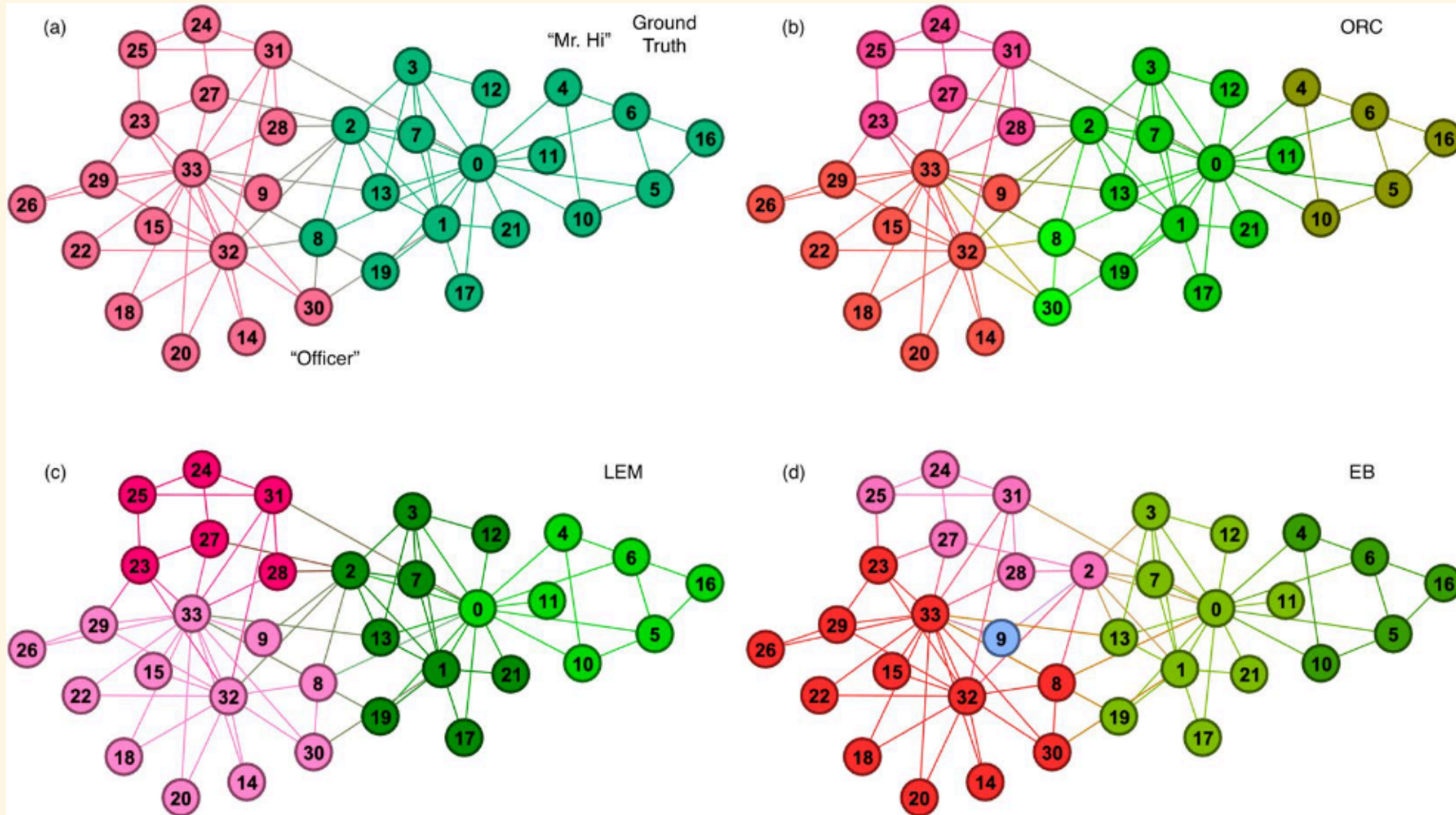
- 4: Clustering
- 5: co-occurrence grouping

For homogeneous networks, eg., country-to-county of Ch. 2

Community: nodes that are closely connected with each other by *strength* or *density*

Resolution limit: communities with less than $\sqrt{|V|}$ members cannot be properly identified.

KARATE CLUB EXAMPLES



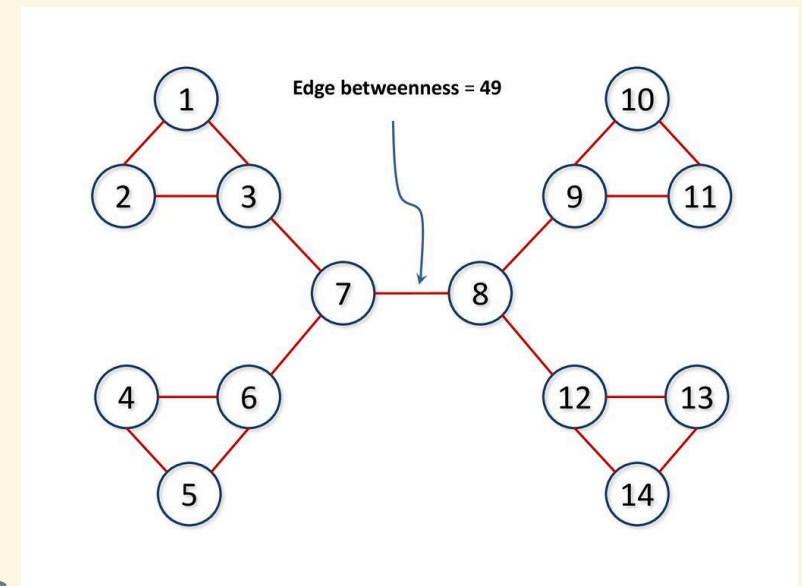
Ground truth vs. community detection in [\[Sia et al., Sc. Rep. 2019\]](#)

GIRVAN-NEWMAN

Hyp: Betweenness centrality captures *help to connectivity*

but removing betweenness we reveal the actual underlying community structure

1. Rank edges by their *help to connectivity*
2. remove the top-ranking edge
3. repeat until loss of connection
4. now-isolated areas are called communities



MODULARITY

AS AN OPTIMIZATION PROB.

Instance: an adj. matrix A , a small integer g

Solution: a partition of V into g groups

Measure: maximise Q : the overall modularity measure

Interpretation: how likely is a random walker to leave the community?

THE Q FACTOR

Let $E_{g \times g}$ be the cross-group matrix and f_i the sum of col. i .

Electrical conductance:

$$Q = \sum_{i=1}^g e_{ii} - f_i^2$$

Complexity: NP-complete: it won't scale up

Even random networks might exhibit 'densifications' that might look as communities.